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Problem Set 1

Autoregressive Process $\sim$ AR(1).
The following process is assumed for this problem set:

$$z_t = \rho z_{t-1} + \epsilon_t \quad \text{where} \quad \epsilon \sim N(0, \sigma^2)$$

This process is going to be discretized through **Tauchen's** and **Rouwenhorst's** method. The following parameter values are considered.

Calibration.

Parameter	Value	Description
ρ	0.95 and 0.99	Persistence of the AR(1) process
σ	0.007	Standard deviation
N	9	Number of grid points
m	3.0	Scaling parameter for the grid width
T	10,000	Number of simulation periods

Main functions.

`setPar()`: Defines the parameters and simulation settings, allowing them to be easily changed throughout the code.

`tauchen_function(; parameters)`: Implements Tauchen's method as presented in class. It returns the state grid and transition matrix, stored as `z_tauchen` and `P_tauchen`, which approximate the AR(1) process.

`rouwenhorst_function(; parameters)`: Implements Rouwenhorst's method as presented in class, recursively constructing the transition matrix. It returns the state grid and transition matrix, stored as `z_rouwenhorst` and `P_rouwenhorst`, which approximate the AR(1) process.

`AR1_process(; ε, parameters)`: Simulates the original AR(1) process using the supplied parameters and normally distributed innovations.

`AR1_discretized(method; parameters, uniforms)`: Simulates the Markov chain obtained using either Tauchen's or Rouwenhorst's method. Both simulations use the same uniform draws, derived from the innovations supplied to the original AR(1) process.

`estimate_rho(z)`: Estimates an AR(1) regression with an intercept using the simulated series and returns the estimated persistence coefficient, $\widehat{\rho}$.

1. Discretize the process using the Tauchen (1986) method. Use 9 points.

$$z_{\text{Tauchen}} \approx \begin{bmatrix} \mathbf{-0.067} & \mathbf{-0.050} & \mathbf{-0.034} & \mathbf{-0.017} & 0.000 & \mathbf{0.017} & \mathbf{0.034} & \mathbf{0.050} & \mathbf{0.067} \end{bmatrix}.$$

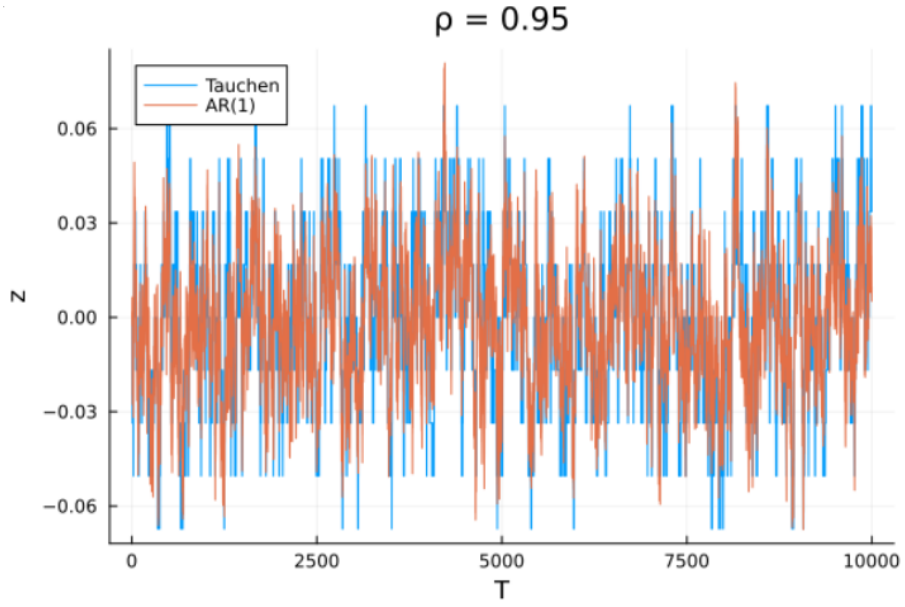
$$P_{\text{Tauchen}} \approx \begin{bmatrix} \mathbf{0.764} & \mathbf{0.235} & \mathbf{0.001} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ \mathbf{0.059} & \mathbf{0.741} & \mathbf{0.200} & \mathbf{0.001} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & \mathbf{0.075} & \mathbf{0.757} & \mathbf{0.168} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & \mathbf{0.093} & \mathbf{0.767} & \mathbf{0.140} & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & \mathbf{0.115} & \mathbf{0.770} & \mathbf{0.115} & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.140} & \mathbf{0.767} & \mathbf{0.093} & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.168} & \mathbf{0.757} & \mathbf{0.075} & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.001} & \mathbf{0.200} & \mathbf{0.741} & \mathbf{0.059} \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.001} & \mathbf{0.235} & \mathbf{0.764} \end{bmatrix}.$$

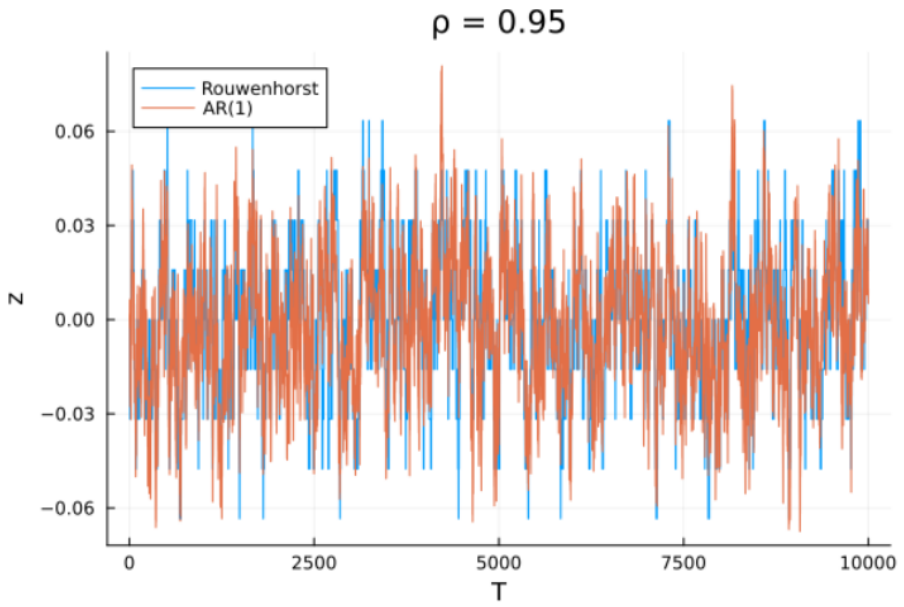
2. Discretize the process using the Rouwenhorst method. Use 9 points.

$$z_{\text{Rouwenhorst}} \approx \begin{bmatrix} \mathbf{-0.063} & \mathbf{-0.048} & \mathbf{-0.032} & \mathbf{-0.016} & 0.000 & \mathbf{0.016} & \mathbf{0.032} & \mathbf{0.048} & \mathbf{0.063} \end{bmatrix}.$$

$$P_{\text{Rouwenhorst}} \approx \begin{bmatrix} \mathbf{0.817} & \mathbf{0.168} & \mathbf{0.015} & \mathbf{0.001} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ \mathbf{0.021} & \mathbf{0.820} & \mathbf{0.147} & \mathbf{0.011} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ \mathbf{0.001} & \mathbf{0.042} & \mathbf{0.823} & \mathbf{0.126} & \mathbf{0.008} & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & \mathbf{0.002} & \mathbf{0.063} & \mathbf{0.825} & \mathbf{0.105} & \mathbf{0.005} & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & \mathbf{0.003} & \mathbf{0.084} & \mathbf{0.825} & \mathbf{0.084} & \mathbf{0.003} & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & \mathbf{0.005} & \mathbf{0.105} & \mathbf{0.825} & \mathbf{0.063} & \mathbf{0.002} & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.008} & \mathbf{0.126} & \mathbf{0.823} & \mathbf{0.042} & \mathbf{0.001} \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.011} & \mathbf{0.147} & \mathbf{0.820} & \mathbf{0.021} \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.001} & \mathbf{0.015} & \mathbf{0.168} & \mathbf{0.817} \end{bmatrix}.$$

3. Simulate the continuous process for 10,000 periods. Do the same for the discretized processes (remember to use the same realizations for the shock for all simulations). Compare the paths of each process (graphs are useful here). If they're not very close, use more points.





4. Estimate the AR(1) processes based on simulated data, both for Tauchen and Rouwenhorst. How close are they from the real data generating process? If they're not very close, use more points.

To assess how closely each simulated series reproduces the original AR(1), I estimated a separate OLS regression for each series:

$$z_t = \alpha + \widehat{\rho} z_{t-1} + u_t.$$

The function `estimate_rho(z)` returns the estimated coefficient $\widehat{\rho}$ for z_{t-1}. I then calculated the series' mean, standard deviation, and persistence error, $|\widehat{\rho} - \rho|$ for each series.

Table 1: Simulation results

Method	$\mu(z_t)$	$\sigma(z_t)$	$\widehat{\rho}$	$\lvert\widehat{\rho} - \rho\rvert$
AR(1)	-0.001912	0.021841	0.947485	0.002515
Tauchen	-0.001125	0.027277	0.954246	0.004246
Rouwenhorst	-0.000585	0.023605	0.954671	0.004671

Both methods approximate persistence $\rho = 0.95$ reasonably well, with Tauchen performing sligthly better than the Rouwenhorst, with a smaller error. However, Rouwenhorst better reproduces the standard deviation $\sigma(z_t)$ of the original AR(1) process.

5. Redo the discretization with $\rho = 0.99$.

5.1. *Discretize the process using the Tauchen (1986) method. Use 9 points.*

Tauchen's method.

$$z_{\text{Tauchen},\,0.99} \approx \begin{bmatrix} \mathbf{-0.149} & \mathbf{-0.112} & \mathbf{-0.074} & \mathbf{-0.037} & 0.000 & \mathbf{0.037} & \mathbf{0.074} & \mathbf{0.112} & \mathbf{0.149} \end{bmatrix}.$$

$$P_{\text{Tauchen},\,0.99} \approx \begin{bmatrix} \mathbf{0.993} & \mathbf{0.007} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ \mathbf{0.002} & \mathbf{0.991} & \mathbf{0.006} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & \mathbf{0.003} & \mathbf{0.992} & \mathbf{0.005} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & \mathbf{0.003} & \mathbf{0.992} & \mathbf{0.005} & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & \mathbf{0.004} & \mathbf{0.992} & \mathbf{0.004} & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.005} & \mathbf{0.992} & \mathbf{0.003} & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.005} & \mathbf{0.992} & \mathbf{0.003} & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.006} & \mathbf{0.991} & \mathbf{0.002} \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.007} & \mathbf{0.993} \end{bmatrix}.$$

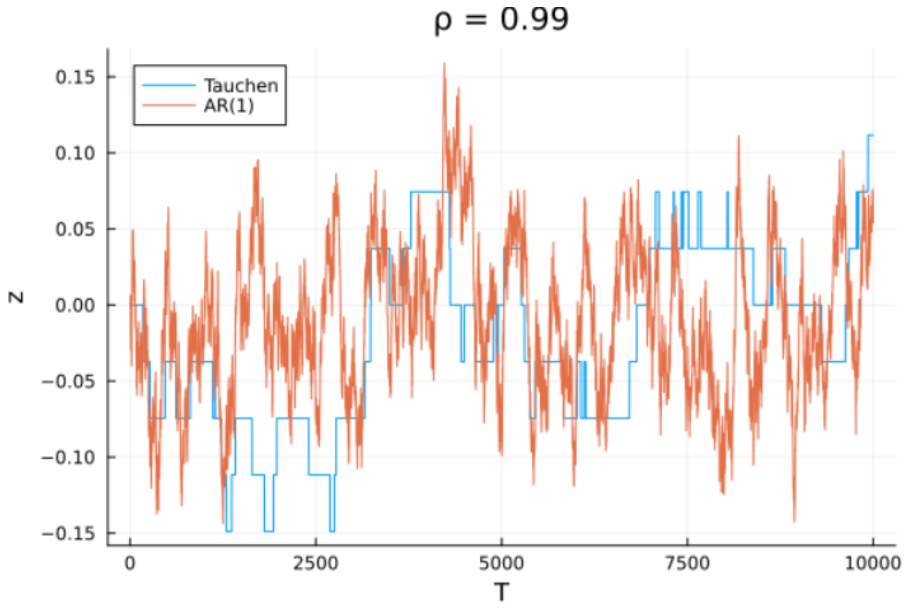
5.2. *Discretize the process using the Rouwenhorst method. Use 9 points*

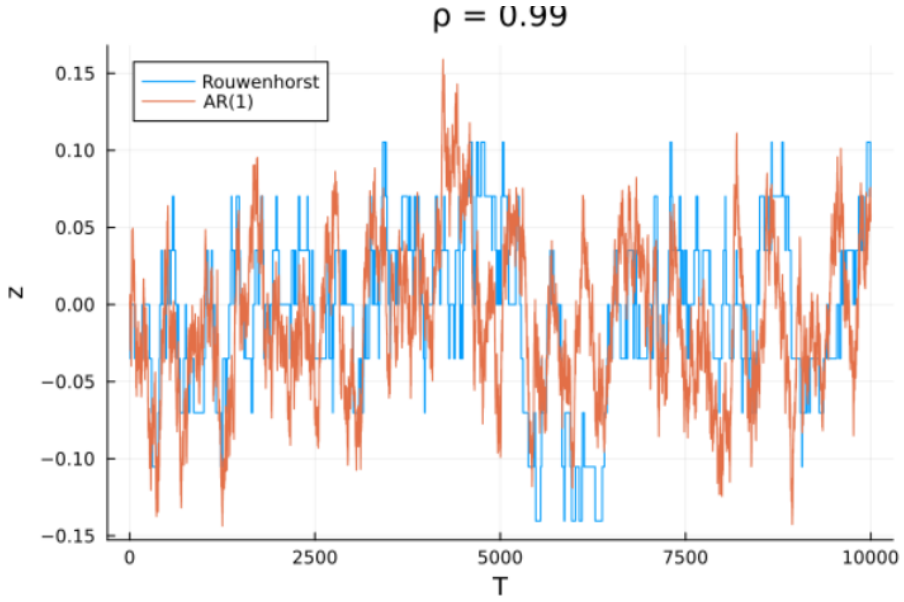
Rouwenhorst's method.

$$z_{\text{Rouwenhorst},\,0.99} \approx \begin{bmatrix} \mathbf{-0.140} & \mathbf{-0.105} & \mathbf{-0.070} & \mathbf{-0.035} & 0.000 & \mathbf{0.035} & \mathbf{0.070} & \mathbf{0.105} & \mathbf{0.140} \end{bmatrix}.$$

$$P_{\text{Rouwenhorst},\,0.99} \approx \begin{bmatrix} \mathbf{0.961} & \mathbf{0.039} & \mathbf{0.001} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ \mathbf{0.005} & \mathbf{0.961} & \mathbf{0.034} & \mathbf{0.001} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & \mathbf{0.010} & \mathbf{0.961} & \mathbf{0.029} & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & \mathbf{0.014} & \mathbf{0.961} & \mathbf{0.024} & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & \mathbf{0.019} & \mathbf{0.961} & \mathbf{0.019} & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.024} & \mathbf{0.961} & \mathbf{0.014} & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.029} & \mathbf{0.961} & \mathbf{0.010} & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.001} & \mathbf{0.034} & \mathbf{0.961} & \mathbf{0.005} \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & \mathbf{0.001} & \mathbf{0.039} & \mathbf{0.961} \end{bmatrix}.$$

5.3. *Simulate the continuous process for 10000 periods...*





5.4. *Estimate the AR(1) processes based on simulated data, both for...*

Table 2: Simulation results, $\rho = 0.99$

Method	$\mu(z_t)$	$\sigma(z_t)$	$\widehat{\rho}$	$\lvert\widehat{\rho} - \rho\rvert$
AR(1)	-0.010042	0.049533	0.990096	0.000096
Tauchen	-0.020953	0.058082	0.998796	0.008796
Rouwenhorst	-0.004649	0.053639	0.991665	0.001665

When persistence increases from $\rho = 0.95$ to $\rho = 0.99$, Rouwenhorst reproduces both the true persistence and the theoretical standard deviation of z_t more acurately than the Tauchen method. The difference is also visible in the graphs. With nine states, Tauchen overstates persistence and dispersion, suggesting that its grid should be refined.